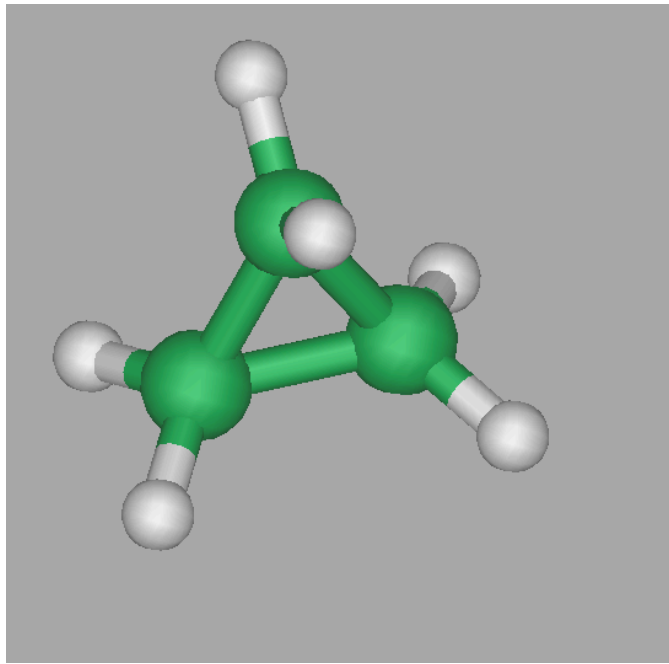


Quantum Mechanics and The Structure of Matter



What do we need to know for thermodynamics?

- How atoms and molecules store energy:
 - Translation, rotation, vibration, electronic motion
- The details of their structure:
 - Quantum mechanics: quantum numbers and properties
- How large numbers behave in equilibrium:
 - Statistics
- Today we will talk about Quantum Mechanics

The Structure of Atoms and Molecules

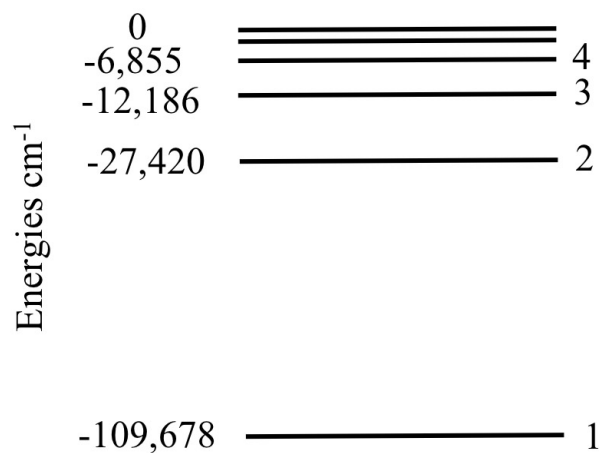
- We know that matter is made up of atoms and that atoms can combine to form molecules
- Atoms are composed of a nucleus and electrons.
- We use the term “structure” to describe the allowed configuration of an atom or molecule.

The Structure of Atoms and Molecules

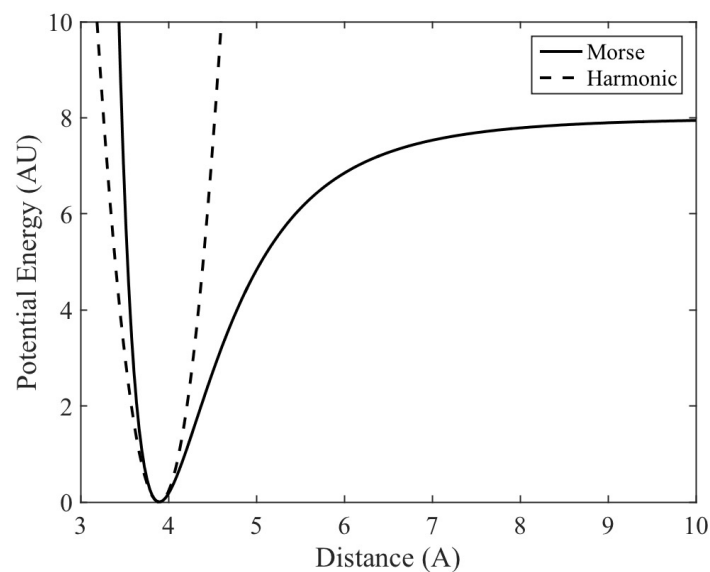
- We know that atoms and molecules generally exist in fixed states, known as quantum states.
- A quantum state is an allowed configuration that determines spatial and dynamic behavior of nuclei and electrons.
- In thermodynamics we are mainly interested in the kinetic and potential energy of the particle.

Energy Level Diagrams

We thus typically describe the structure using energy level diagrams



Hydrogen Atom



Diatomic Molecule

Quantum Mechanics

- Quantum mechanics is the mathematics of atomic and molecular structure.
- We seek to determine allowed quantum states and their energies.
- We will next explore a bit of the history of quantum mechanics.